

CoRoT-1b

R-2372

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_220327 · created 2026-08-20T00:28:59+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459665.6706 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.04
Transit depth (Rp/Rs)^2	2.04 +/- 1.13 [%]
Orbital Inclination (inc)	82.79 +/- 2.85
Airmass coefficient 1 (a1)	1.38 +/- 0.044
Airmass coefficient 2 (a2)	-0.1252 +/- 0.0069
Scatter in the residuals of the lightcurve fit is	3.09 %
Best Comparison Star	None
Optimal Aperture	3.05
Optimal Annulus	7.55
Transit Duration (day)	0.042 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.04 vs 0.1388 (deviation 0.1σ)
Duration fit vs published	0.042 d vs 0.1046 d (deviation 5.69σ)
Mid-time O–C	-12.7 min
Depth SNR	3.6
Residual scatter	3.09 %

Light curve

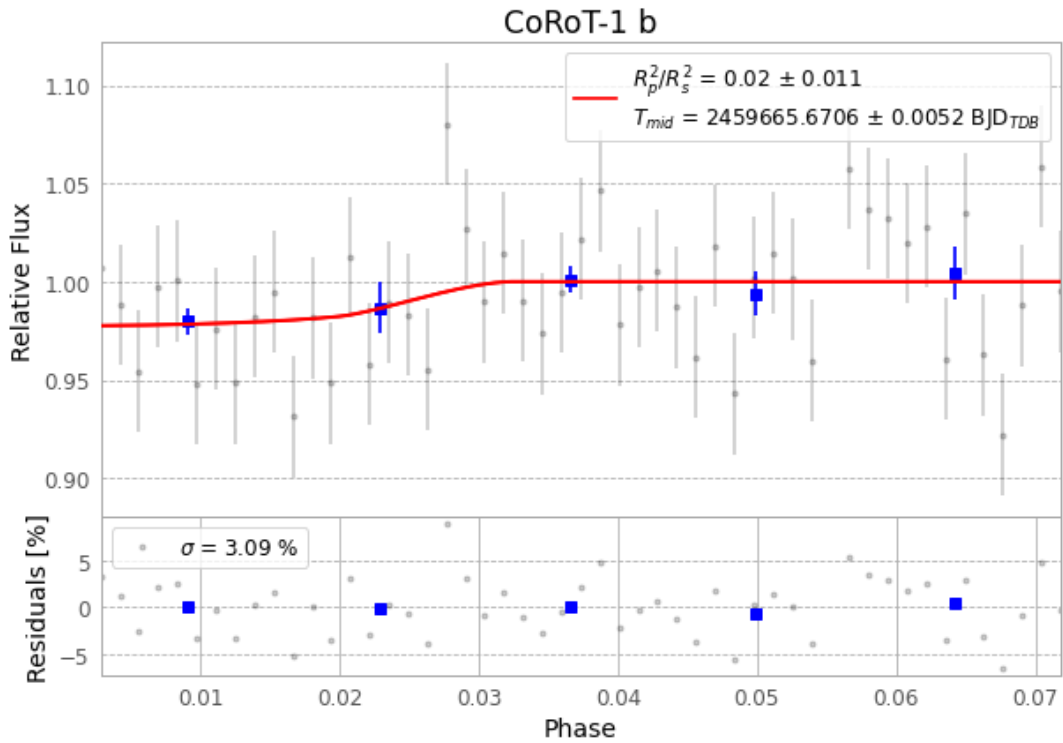


Figure 1 — FinalLightCurve_CoRoT-1 b_2022-03-26.png (SHA-256 f59d8ee5639ff45c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [226, 205], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 ab83e3bd27567c74...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

51 FITS frames (33 MB), CoRoT-1220327040912.FITS → CoRoT-1220327063913.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-220327.log (42436a83aef0...), FinalLightCurve_CoRoT-1 b_2022-03-26.png (f59d8ee5639f...), FinalLightCurve_CoRoT-1 b_2022-03-26.csv (60d4f0936314...)

Compute resources

Wall time 113.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 00:29Z.